

Leela Prashanthi Surisetty

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Profile

AI/ML Engineer with hands-on experience in developing real-time facial recognition systems using YOLOv8, OpenCV, and Python. Skilled in machine learning, computer vision, data preprocessing, and model optimization, with experience building practical AI solutions focused on accuracy and real-world implementation. Proficient in Python, Java, Pandas, NumPy, and OpenCV through academic projects and internship-based applications. Experienced in applying machine learning techniques to solve data-driven problems and improve system performance.

Education

B.Tech CSE (AI/ML) GITAM University, Visakhapatnam CGPA: 8.61	2023 – 2027
Intermediate (MPC) Sri Chaitanya, Visakhapatnam Percentage: 90.4%	2021 – 2023

Projects

Facial Recognition System using YOLOv8 and OpenCV

- Developed a real-time facial recognition system using YOLOv8, OpenCV, and Python for face detection and identification
- Trained the model on 50,000+ facial images and improved detection accuracy from 82% to 87%
- Implemented real-time face detection and recognition pipeline for efficient identity verification
- Integrated Telegram alert notifications to detect and report unknown individuals for enhanced monitoring and security

Premium Eats - Food Ordering Web Application [Live Demo]

- Developed a responsive food ordering platform with user authentication, cart management, and order processing features
- Designed backend functionalities using Java Servlets and MySQL for efficient data and order management
- Implemented dynamic frontend interactions using JavaScript for real-time cart updates and improved user experience
- Managed project deployment and version control using GitHub and Git workflows

Research Work

Facial Recognition Performance Analysis using Deep Learning

May 2026 – Present

- Conducting comparative research on face detection and recognition models including YOLOv8, MTCNN, FaceNet, ArcFace, and Buffalo_L.
- Evaluating model performance on real-world datasets under varying conditions such as illumination changes, pose variations, and facial occlusions.
- Analyzing recognition accuracy, precision, recall, F1-score, and inference time for different model combinations.
- Investigating the effectiveness of hybrid pipelines including YOLOv8 + FaceNet, YOLOv8 + ArcFace, MTCNN + FaceNet, and MTCNN + ArcFace.
- Preparing a research paper focused on enhancing facial recognition performance for surveillance, security, and identity verification applications.

Experience

Machine Learning Intern | Code Beat

May 2024 – Jul 2024

- Built and optimized machine learning pipelines on real-world datasets
- Applied supervised and unsupervised learning algorithms for predictive analysis
- Performed data preprocessing, feature engineering, and model evaluation
- Evaluated model performance using accuracy metrics and confusion matrices

Technical Skills

- **Languages:** Python, Java, SQL, C
- **AI/ML:** Machine Learning, Computer Vision, Data Preprocessing, Model Optimization
- **Libraries & Frameworks:** YOLOv8, OpenCV, Pandas, NumPy, Scikit-learn
- **Tools:** Git, GitHub, Google Colab
- **Core Concepts:** Data Structures, OOP, DBMS

Certifications

- Data Analysis with Python — **FreeCodeCamp**
- AWS Solutions Architect — In Progress